Very predictable Exam One: Nov 5th in class.

Tanks, linear first-order ODEs, separable ODEs, and first-order constant-coefficient linear systems. All including y_c, y_p, and initial conditions.

- Four questions on two sheets totaling 50 points.
- Questions like HW with fewer boxes stuck together.
- Write anything you want on a two-sided notecard less than 4×6 inches.
- Almost all review this week.

Separable and Linear ODEs

Separable $y' = g(t)h(y)$

- Multiply by dt and separate variables $\frac{dy}{h(y)} = g(t)\,dt$
- Integrate, then use the initial condition to determine C.
- Do not solve for y on an exam.

Linear $a_1(t)y' + a_0(t)y = f(t) \implies y' + p(t)y = q(t)$

- Divide by $a_1(t)$ to get the standard form $y' + p(t)y = q(t)$
- Multiply by the integrating factor $\mu(t) = \exp\left(\int p(t)\,dt\right)$ to get $(\mu y)' = \mu q(t)$.
- Integrate then use the initial condition to determine C.

Linear Systems $y' = Ay + f$: $y = y_c + y_p$ for $A \in \mathbb{R}^{n \times n}$

Complementary solution $y_c' = Ay_c$

- Find n linearly independent real solutions $\boldsymbol{y}_k(t)$ from eigenvalues and eigenvectors.
- Real eigenpairs give $e^{\lambda t}\boldsymbol{v}$. Splitting complex eigenpairs gives two LI solutions. Then $\boldsymbol{y}_c = \sum_{k=1}^{n} c_k \boldsymbol{y}_k$.

Particular solution $y_p' = Ay_p + f(t)$

- Substitute the undetermined coefficient form matching $\boldsymbol{f}$ and match coefficients.
- Add particular solutions for separate forcing terms.

For $\boldsymbol{y}(0) = \boldsymbol{y}_0$, determine $\boldsymbol{c}$ by solving $\boldsymbol{Mc} = \boldsymbol{y}_0 - \boldsymbol{y}_p(0)$, where $\boldsymbol{M} = [\boldsymbol{y}_1(0) \ \cdots \ \boldsymbol{y}_n(0)]$.